Effect of extracted housefly pupae peptide mixture on chilled pork preservation.

The peptide mixture from housefly pupae has broad spectrum antimicrobial activity but has not previously been reported as a food preservative. In this study, the preservation effects of a housefly pupae peptide mixture, nisin, and sodium dehydroacetate (DHA-S) on the number of mesophilic aerobic bacteria (MAB), total volatile basic nitrogen (TVB-N), and pH value of chilled pork were compared. All results showed that a good preservation effect was observed among 3 treatments with the peptide mixture of housefly pupae, nisin, and DHA-S and that there was no significant difference among them. These results indicate that housefly peptide mixture has a great potential as a food preservative. The results of scanning electron microscope and transmission electron microscopy suggest that the primary mechanism of housefly pupae peptide mixture may be bacterial cytoplasmic membrane lysis and pores induced in the membranes. Practical Applications: Peptide mixture extracted from housefly pupae using low-cost and simple method has broad spectrum antimicrobial activity. According to the effect on chilled pork preservation, extracted housefly peptide mixture has a great potential as a food preservative.